UNIVERSITÀ DI PARMA

The future of xAI

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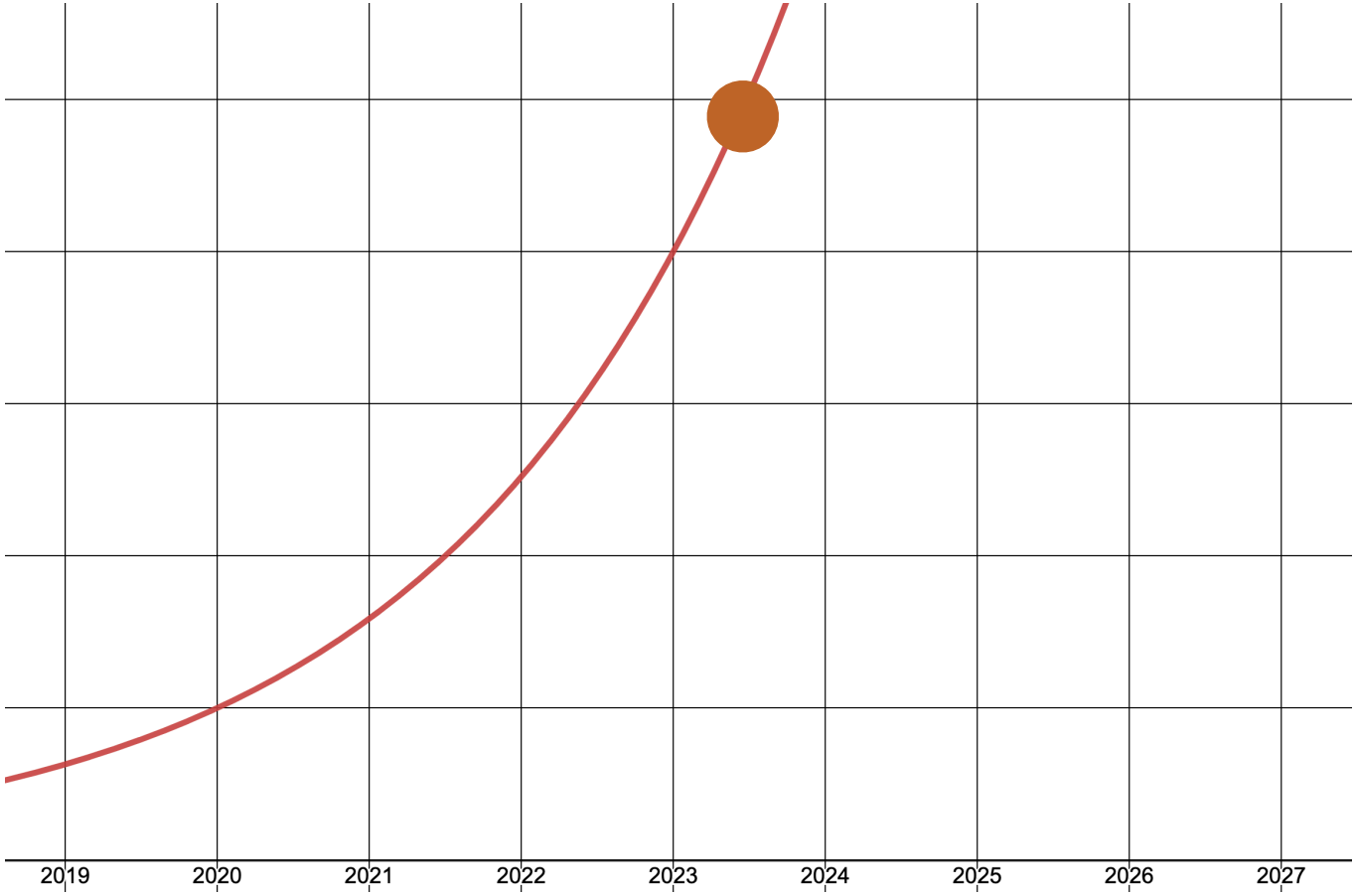
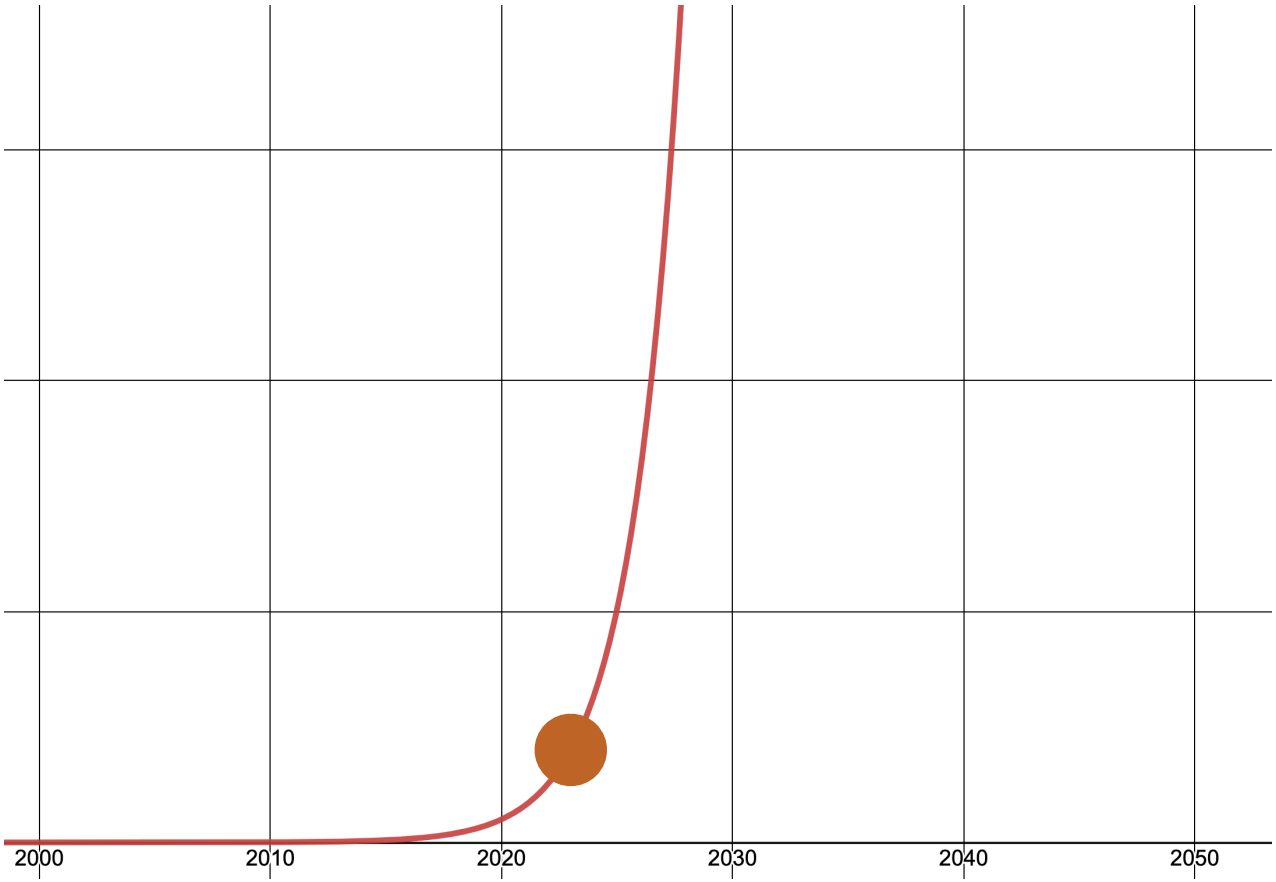
An exponential change

- CPUs progresses (Moore's law since 1965 -> 2x every 1.5 years)
- Computing resources -> AI capability of learning



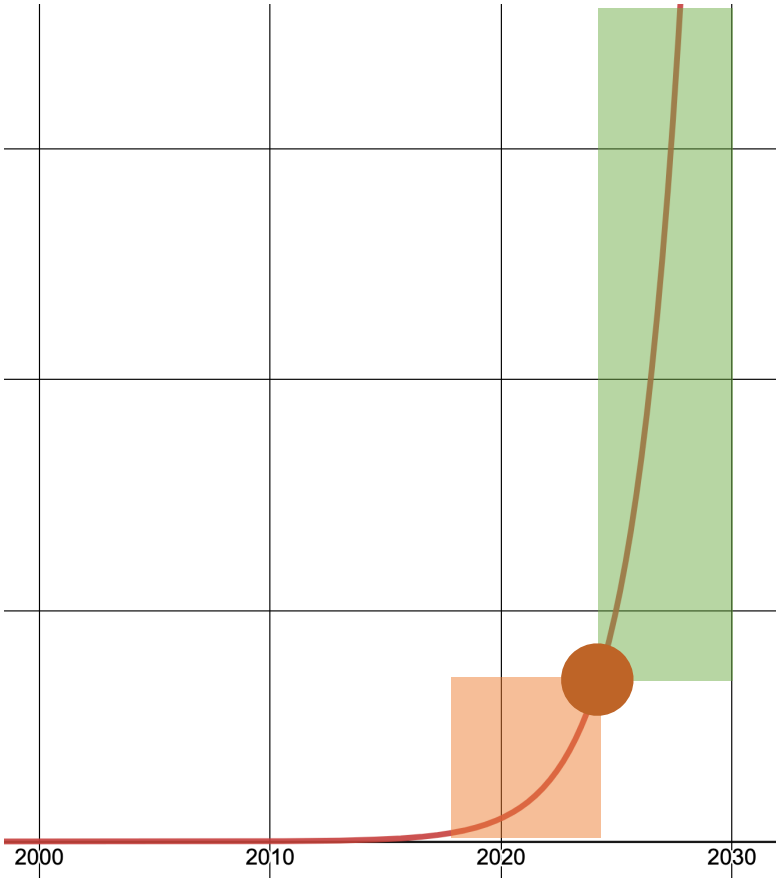
An exponential change

- Human evolution is not suitable for adapting to exponential phenomena
- Are able to understand, control and predict AI?



A milestone for AI: EU AI ACT

- Evolution 2018-2023 —> EU AI ACT
- It rules the world 2024—… completely different from the past
- Guidelines are necessary
- First worldwide law to rule IA with a comprehensive scope

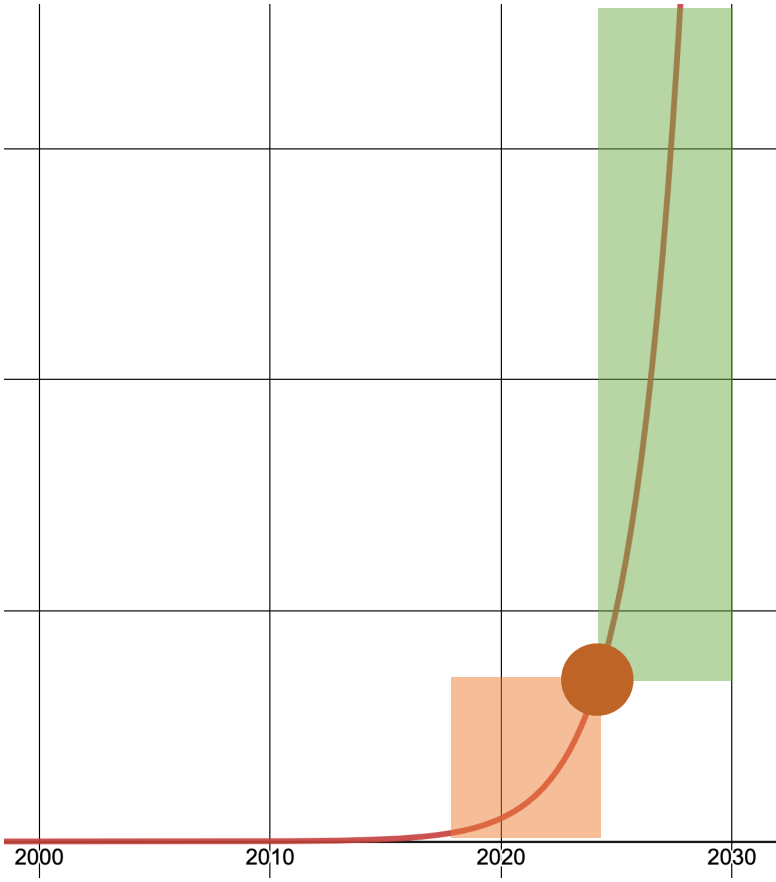


A milestone for AI: EU AI ACT

https://www.agendadigitale.eu/cultura-digitale/artificial-intelligence-act-a-che-punto-siamo-e-i-principali-nodi-da-sciogliere/

1. AI definition
2. Forbidden activities
   - Social scoring
   - Remote biometrics
   - Subliminal methods / fragile people
3. Risk based classification of AI:
   - Medical, legal applications
   - Scoring (students, financial, CV)

} High Risk



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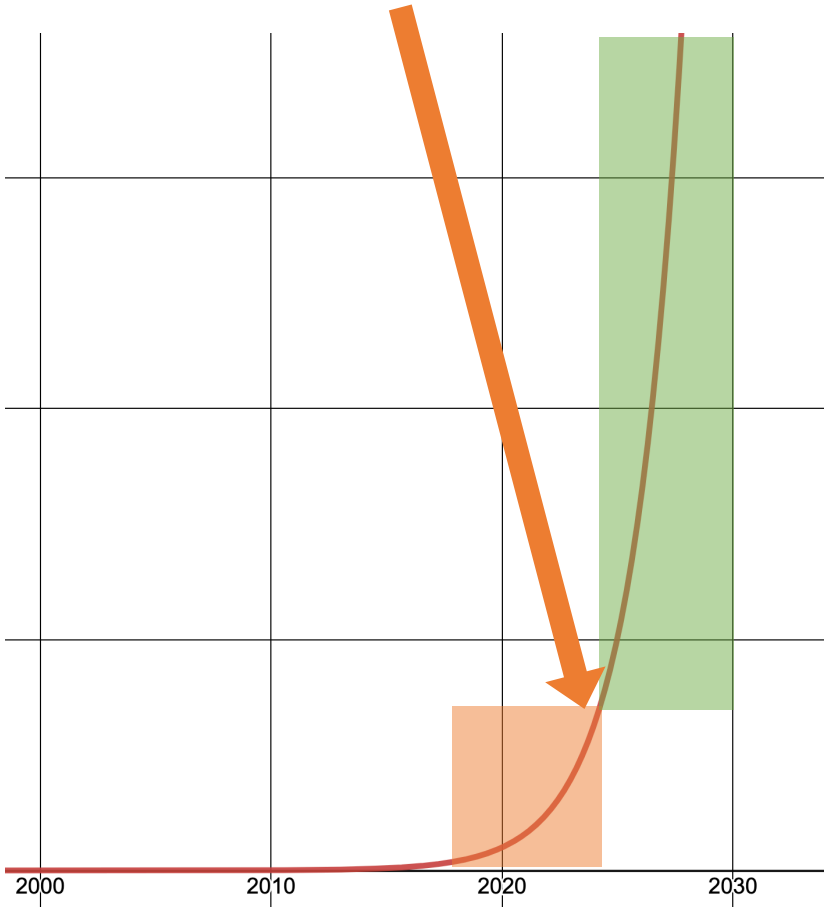
- Generative AI not explicitly included in AI Act

  (ChatGPT, Dall-E, Midjourney, …)

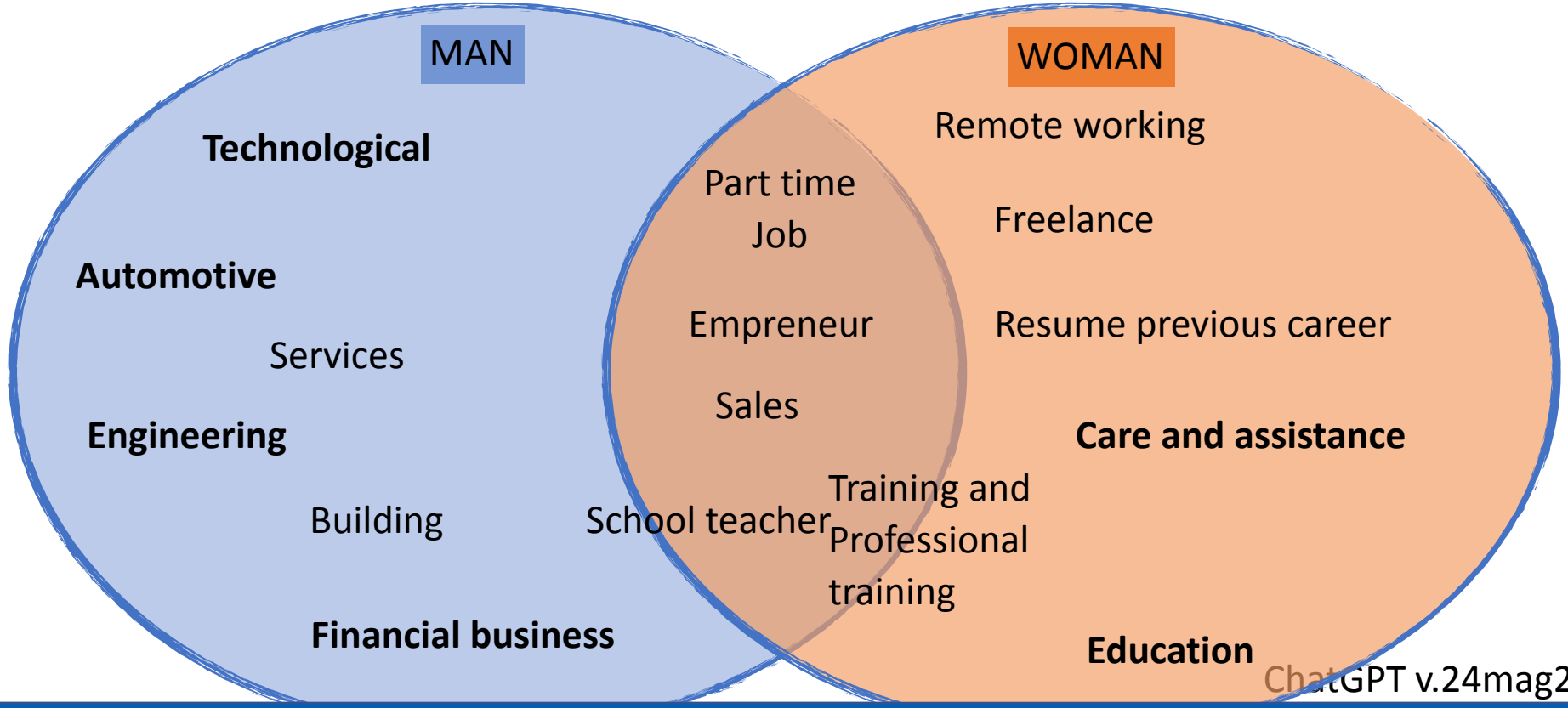
- New issues:
  - Copyright (✓ Firefly Adobe (may 23))
  - Bulk data retrieval
  - Legal liability for generated output
  - Embedding into other services (API):
    - GPT4 -> Bing (mar 23)



Generative AI risks

Bias: AI learns from **unbalances** datasets > this is our society -> discriminations

> I'm a man/woman, what job I should start?



ChatGPT v.24mag23

Generative AI risks

- Blackbox:
  - Algorithms / training dataset **not available** (e.g. ChatGPT)
  - Can’t analyze the trained network (propietary)
  - Even if available: trilions of parameters!

- Improper usage: AI for a specific and unruled task (e.g. fake news)

Explainable AI

https://ec.europa.eu/futurium/en/system/files/ged/ai-and-interpretability-policy-briefing_creative_commons.pdf

- Are we happy with high accuracy? (Images OK, text?)
- **Ethical principles** to include AI in the decisional chain:
  - System's trust
  - Bias control
  - Compliance to standards/directives
  - Incremental improving
- How?
  - **Explainability**: allow to understand how/why I get that answer
  - **Transparency and interpretability**: I can access and understand algorithms and data
  - **Logical proof**: on what basis I get that answer?

Explainable AI

Rudin, Cynthia. "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead." *Nature machine intelligence* 1.5 (2019): 206-215.

- Create new AI to explain what an AI is doing?
- Blackbox to explain blackbox…
- Accuracy gets combines (and decreases)
- Unfeasible target

- Native explainability: a new research direction

Two joining paths?

www.ted.com/talks/gary_marcus_the_urgent_risks_of_runaway_ai_and_what_to_do_about_them/c

Machine learning
Generative AI

- "Learns" easily from data
- Does not abstract data
- No programming
- **Blackbox**
- Billions of data handled
- Training is expensive

Symbolic AI

- Knowledge representation
- Automated reasoning
- Programming (e.g. Constraint!)
- **Natively explainable**
- Computation expensive

Two joining paths?

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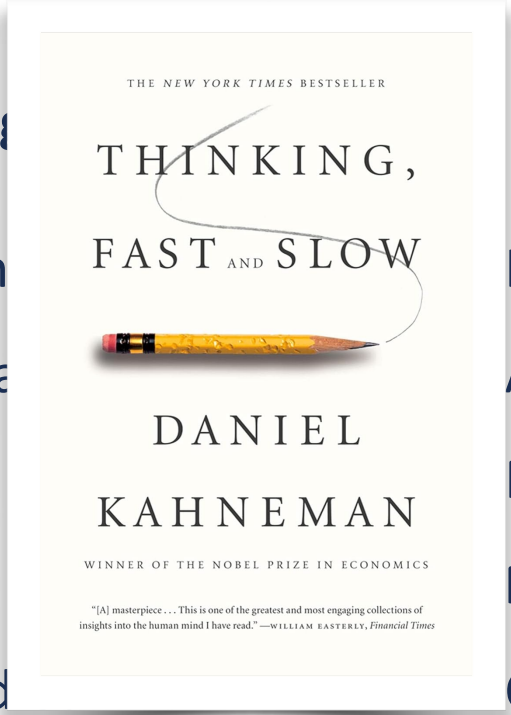

Symbolic AI

- Knowledge representation
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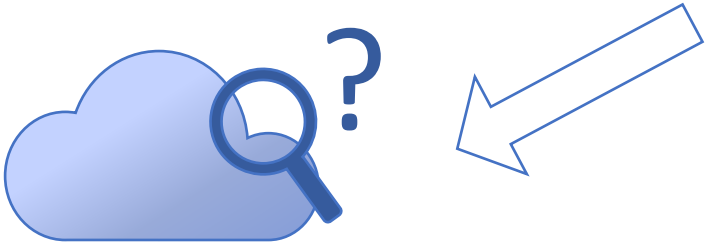
?

Two joining paths?

Machine learning
ChatGPT

- Syntax level
- Probability for next token
- Typical NL expression patterns
- It somewhat captures a certain semantics (which one?)

Symbolic AI

- Semantic level
- Relations on objects
- Similar to the human deductions
- **Can it explain ML?**